

Complain like you mean it! How prosody conveys suffering even about innocuous events

Maël Mauchand^{*}, Marc D. Pell

McGill University, School of Communication Sciences and Disorders, Montréal, Québec, Canada

ABSTRACT

When complaining, speakers can use their voice to convey a feeling of pain, even when describing innocuous events. Rapid detection of emotive and identity features of the voice may constrain how the semantic content of complaints is processed, as indexed by N400 and P600 effects evoked by the final, pain-related word. Twenty-six participants listened to statements describing painful and innocuous events expressed in a neutral or complaining voice, produced by ingroup and outgroup accented speakers. Participants evaluated how hurt the speaker felt under EEG monitoring. Principal Component Analysis of Event-Related Potentials from the final word onset demonstrated N400 and P600 increases when complainers described innocuous vs. painful events in a neutral voice, but these effects were altered when utterances were expressed in a complaining voice. Independent of prosody, N400 amplitudes increased for complaints spoken in outgroup vs. ingroup accents. Results demonstrate that prosody and accent constrain the processing of spoken complaints as proposed in a parallel-constraint-satisfaction model.

1. Introduction

In our everyday use of language, the implications of a conversation often go beyond the actual topic being discussed. During such interactions, the verbal information being conveyed may largely provide instrumental support to achieve more basic social goals of the speaker that are signaled paralinguistically - for instance, vocal cues that foster interpersonal affiliation during speech communication (Scarantino, 2017; M. Tomasello, 2008).

A prime example of such “non-instrumental” speech acts is third-party complaining. A third-party complaint is a verbal report of a distressing, painful, or annoying event suffered by the speaker, which is communicated to a listener with no agency in that event (Mauchand & Pell, 2021; Ogden, 2007). Importantly, complaints are usually not focused on the event itself, but rather on the subjective state of the speaker; what the complaint is about, and what can be done about it, are less important than how the speaker *feels* (Drew & Walker, 2009; Edwards, 2005). Complaining exchanges often drift away from their initial topic, but always remain centered on their purpose of affect sharing between the complainer and complaine. In fact, many complaints do not relate “true” suffering but describe harmless or innocuous events of little importance (Alicke et al., 1992; Boxer, 1993). While eliciting empathy and affiliation by describing low-suffering situations may seem counter-intuitive, this communicative strategy effectively facilitates the empathic involvement of listeners in different types of social contexts, from routine rituals among intimates to ice-breaker

conversations between strangers (Boxer, 1993; Kowalski, 2002).

Given that the events described in complaints are not always inherently painful, complaining often functions as an emotive act: a social performance in which affect is not directly experienced but reconstructed (Caffi & Janney, 1994; Selting, 1994, 2010). In order for discourse to efficiently convey emotivity without salient verbal cues, speakers often modulate their vocal expression (*prosody*) to signal their affective disposition (Acuña-Ferreira, 2002; Ogden, 2010). Prosody is a powerful medium to convey emotions and can act as a negotiating device in social interactions (Pell & Kotz, 2021; Szczeppek Reed, 2011; Truesdale & Pell, 2018). As a listener, the prosodic form of an utterance may facilitate third-person appraisal: reasoning forward about how an event may affect the subjective state of the speaker (Ong et al., 2018). As such, prosody can increase listener motivation, persuasion, and involvement even in low-involvement contexts (Gelinas-Chebat & Chebat, 1992; Van Zant & Berger, 2020; Zougkou et al., 2017).

When complaining, prosody serves as the main emotive device to elicit empathy in the listener through acoustic markers of affect and expressivity, such as heightened pitch, greater pitch variability, and rhythmic emphasis (Acuña-Ferreira, 2002; Ogden, 2007; Rao, 2013). Recent acoustic and perceptual analyses suggest that complaints exhibit changes in voice quality and high-frequency energy similar to pain vocalizations (Mauchand & Pell, in review; Raine et al., 2019). Complaining prosody is easily discriminated from neutral speech, and perceptually associated with negative, high-arousal emotions such as anger, sadness, and surprise (Mauchand & Pell, 2021, in review). The

^{*} Corresponding author at: 2001 McGill College - Rm. 850, Montréal QC H3A 1G1, Canada.
E-mail address: mael.mauchand@mail.mcgill.ca (M. Mauchand).

characteristic “sound of complaints” serves a dual social function to implicitly increase affect sharing and to explicitly signal the emotive, complaining nature of the current discourse. In so doing, the speaker ensures the empathic involvement of the listener (Selting, 2010); even when they describe seemingly innocuous events, prosody may thus point the listener towards a painful, suffering interpretation of their utterance and promote empathic affiliation.

How, then, does a speaker's prosody affect how listeners process complaining statements which do not always describe emotionally painful events? While the meaning of spoken utterances unfolds progressively, there is evidence that listeners use currently available cues and knowledge sources interactively to make predictions about upcoming words in a goal-directed fashion as an utterance representation is formed (Kuperberg & Jaeger, 2016; Kutas & Federmeier, 2010). Neurophysiological tools such as electroencephalography (EEG) and event-related potentials (ERPs) provide insight as to how predictions derived from vocal speech cues facilitate, or fail to facilitate, the semantic processing of linguistic cues at specific points of an utterance (Van Berkum et al., 2008). Pragmatic processing of speech signals occurs almost immediately from utterance onset, indexed by early ERP components such as the N100 and P200 (Mauchand & Pell, 2022b; Paulmann et al., 2011; Pell et al., 2015), and continues as speech unfolds through the on-line integration and (re)interpretation of novel information (see Pell & Kotz, 2021; R. Tomasello, 2023 for reviews). In particular, N400 and P600 components, often modulated by semantic and pragmatic manipulation, are thought to index efforts in word meaning retrieval and in-utterance word integration, respectively (Aurnhammer et al., 2021; Brouwer et al., 2017; Delogu et al., 2019; Van Petten & Luka, 2012).

While most early evidence for N400 and P600 responses results from sentences manipulated to contain anomalous, mismatching, or unexpected words (Kutas & Federmeier, 2010; Lau et al., 2008), recent research reports modulation of these semantic components as listeners process more natural types of discourse, including irony (Caillies et al., 2019; Cornejo et al., 2007), world knowledge and (dis)trust (Foucort & Hartsuiker, 2021; Jiang et al., 2020), imposing requests (Vergis et al., 2020), or innuendo (Rigoulot et al., 2020). In these cases, increased N400 (and sometimes P600) amplitudes were evoked by mismatches between the current word and contextual cues (e.g., irony, truth value), task demands (e.g., assess confidence, friendliness), or general conversational rules (politeness, appropriateness). While in these diverse and complex social-pragmatic tasks the roles attributed to the N400 and P600 vary, N400 responses usually pertain to the relationship between the current word and other lexical, vocal, or task constraints (Molinaro et al., 2009; Paulmann & Kotz, 2008; van den Brink et al., 2012), whereas P600 effects indicate pragmatic integration efforts of the whole utterance within its context (Regel et al., 2011; Rigoulot et al., 2014).

The sensitivity of word and utterance meaning retrieval to such a wide variety of factors reveals the complex set of constraints that can simultaneously impose on speech processing. To account for the integration of all this information, parallel constraint-satisfaction models suggest that listeners process social and stereotyping signals immediately as they are available to form impressions about speakers and to constrain the interpretation of discourse (Kunda & Thagard, 1996; Pexman, 2008). New signals that do not satisfy these impressions will have increased surprisal and require greater effort to retrieve their meaning, whereas words whose meaning has been pre-activated will be processed more easily (Caffarra et al., 2019; Levy, 2008). In the context of complaint evaluation, the goal of assessing speaker suffering should constrain listener's attention to expect pain-related verbal cues; in this case, statements describing innocuous events would have high surprisal potential and increase processing demands over statements which describe explicit situations involving (social) pain.

However, the speaker's prosody is also a potential constraint which could modulate the surprisal of words in spoken complaints, altering how speaker intentions are inferred (Hellbernd & Sammler, 2016;

Wichmann, 2002, 2000). A recent study by Mauchand et al. (2021) revealed that statements such as, *You are such a great cook*, produced with an obviously ironic tone of voice elicited smaller N400 and P600 responses at the critical word than when spoken in a literal tone. This finding suggests that the prosodic signature of irony/sarcasm registered from the onset of the utterance allowed listeners to better predict the ironic intention of upcoming words than a less marked literal voice. Zhang et al. (2021) also reported that prosodic accentuation (e.g., higher pitch) was associated with N400 reductions in high-surprisal words within discourse. These results are important in light of recent work on complaint perception, which showed that utterances produced in an expressive complaining voice are motivationally more salient and cortically differentiated from utterances produced with neutral prosody as early as 200 ms after utterance onset (Mauchand et al., 2022). It can be hypothesized that early and continuous appraisal of the speaker's emotive prosody would guide the listener's interpretation towards an impression of suffering even in the absence of pain-related verbal information.

Another contextual variable which could impact on how listeners process complaining utterances is the speaker's *identity* based on their speech characteristics (e.g., accent). A speaker's perceived linguistic-cultural background is rapidly inferred from their accent, which can induce biases that alter social interactions (Fiske et al., 2016). Out-group perception is known to activate implicit negative biases that affect downstream semantic processing (Gawronski & Bodenhausen, 2006; Mauchand & Pell, 2022a; Pantos & Perkins, 2012). Important to the processing of complaints, which are designed to promote affect sharing, empathy is highly sensitive to outgroup biases, with out-group perception being associated with altered, often diminished empathic responses (Bestelmeyer et al., 2015; Cheon et al., 2010; Sessa et al., 2014; Xu et al., 2009). In language processing, accented speech has been found to elicit greater N400 responses than ingroup accent overall (Goslin et al., 2012; Romero-Rivas et al., 2015), but can also modulate N400 effect in more complex contexts such as credibility evaluations (Foucort & Hartsuiker, 2021; Jiang et al., 2020) or accent-based stereotype violation (Pélissier & Ferragne, 2021).

Specific to this study, recent data point to important differences in the neurocognitive processing of ingroup vs. outgroup complaints at utterance onset, with out-group complaints being marked as less salient and demanding greater empathic (cognitive) effort to evaluate than ingroup complaints (Mauchand & Pell, 2022). However, it is unclear whether early differences in how listeners appraise ingroup vs. out-group emotive signals based on the speaker's prosody and accent-related cues would impact on late integration stages at the sentence-final word, for example, by promoting shallower pragmatic processing of outgroup complaints with effects on the word-final N400 (Foucort et al., 2019; Foucort & Hartsuiker, 2021). Exploring this hypothesis was a secondary aim of our study.

The present study investigated how vocal-speech cues (prosody, accent) influence and hypothetically *constrain* how listeners appraise speaker suffering during online processing of verbal complaints. Given our participants' instruction to focus on how the complainer was feeling (extent of suffering), we expected that utterances describing innocuous events would be associated with greater lexical retrieval (increased N400) and pragmatic integration (increased P600) efforts on the final word than utterances describing task-relevant, socially painful events. However, given evidence that the neurocognitive system differentiates emotionally expressive and neutral prosody almost immediately following utterance onset (Mauchand & Pell, 2022), we predicted that lexical-semantic processing of the final word would be strongly constrained/facilitated by the complaining, emotive tone of voice, which would mark the utterance as an attempt to induce empathy in the listener even when the subject of the complaint was seemingly innocuous (reducing differences in the N400/P600 following complaining prosody). Given evidence that speaker accents tend to promote a more effortful set of procedures to infer the pragmatic meaning of a speaker

(Jiang et al., 2020; Mauchand & Pell, 2022), we speculated that the ingroup/ outgroup status of the speaker would additionally affect how prosodic information is integrated at the final word, albeit in a less clear manner.

2. Methods

2.1. Participants

A power analysis was run using G*power to determine a minimum sample size for the experiment (Faul et al., 2007). Expecting large effects of prosody, but conservatively assuming medium effect sizes regarding accent effects on the ERP components of interest (Jiang & Pell, 2015; Mauchand et al., 2021), a minimum of 24 participants was determined to achieve a power over 80%. Twenty-eight French participants were recruited in the region of Montreal, Canada. Two participants could not complete the experiment due to technical issues, leaving 26 subjects (16F, 10 M; Age: $M = 26.08$, $SD = 4.09$; years of education: $M = 16.23$, $SD = 2.55$) for the analyses. All participants were born in France, had lived in France until they were at least 18, and were living in Montreal to pursue studies or work. To limit effects of long-term habituation to the regional Québécois accent (Sumner & Samuel, 2009), participants were selected based on having lived less than two years in the province of Québec. All participants were right-handed and reported no history of major psychiatric or neurological illness or speech/hearing problems. Participants voluntarily consented to take part in the study which was ethically approved by the Faculty of Medicine Institutional Review Board (McGill University, Montreal, Canada). Participants were remunerated for their involvement (\$40CAD).

2.2. Stimuli

Stimuli were 320 short French utterances created, validated, and selected in two previous studies focusing on the acoustic and perceptual aspects of complaining speech (Mauchand & Pell, in review, 2021b) and also used in a previous ERP study focusing on the neurocognitive effects of complaining prosody (Mauchand & Pell, 2022). Stimuli were constructed in the form of token sets composed of 4 utterances. Each token set was built from a root sentence describing an action performed by a third party and then distinguished in terms of verbal content and prosody. The verbal content was manipulated by changing only the last word of the statement to describe either a socially harmful event (e.g., “Il a dit que j’étais stupide/He said I was stupid”) or an innocuous event (e.g., “Il a dit que j’étais sorti/He said I was outside”).¹ For each sentence pair, the final key words were matched for syllable length and did not significantly differ in word frequency ($t(35.14) = -1.00$, $p = .326$) based on the French database LEXIQUE (New et al., 2001).

Prosody was then manipulated by having the speakers utter each sentence in a complaining and in a neutral tone of voice. For ecological validity purposes, speakers were required to direct their speech to a third person present in the room (i.e., not the examiner) but were otherwise given minimal instructions/cues on how to express the utterances. For neutral utterances, they were told to simply report the event to the third party; for complaints, they were told to complain as if the event had hurt them. Speakers were free to repeat utterances at will until satisfied; the last repetition was selected. As described in other research using the same stimulus set, prosodic distinctions were acoustically and perceptually detectable from utterance onset (Mauchand & Pell, in review, 2021, 2022b). Compared to neutral, complaints were characterized by significantly increased fundamental frequency mean and variability (see Appendix Figure A1) as well as changes in voice quality parameters (shimmer, HNR, high-frequency energy). After performing the prosody manipulation, each token set was thus comprised of

four distinct combinations of prosody/statement attributes: *Neutral/Pain*, *Neutral/Innocuous*, *Complaint/Pain*, *Complaint/Innocuous*.

All utterances were produced by 4 French and 4 Québécois speakers to further modulate the speaker’s accent. From the perspective of our participants, the French accent would lead to an “ingroup” categorization, whereas the Québécois accent would promote an “outgroup” categorization (Mauchand & Pell, 2022a). Recordings were digitally captured in a sound-attenuated chamber with a high-quality head-mounted microphone onto a Tascam recorder (sampling rate of 44.1 kHz, 16-bit, mono, .wav format). They were then edited in Praat (Boersma & van Heuven, 2001) into short.wav audio files and normalized to a peak intensity of 70 dB. There were 672 utterances produced in the initial recording; following a validation study, 10 token sets were selected for each speaker such that each token set was present only once in both accents (Mauchand & Pell, 2021). In summary, there was a total of 2 Accents $\times$ 4 Speakers $\times$ 10 Token sets $\times$ 2 Statements $\times$ 2 Prosodies = 320 utterances.

2.3. Procedure

During the EEG task, participants were seated comfortably in an electrically shielded, sound-attenuating booth. Stimuli were presented through earphones in a pseudo-randomized order that prevented the direct repetition of utterances from the same speaker or token set. In each trial, an utterance was presented after a jittered fixation point (500 to 1500 ms) followed by a question prompting them to evaluate “how hurt the speaker feels” (“À quel point cette personne se sent-elle blessée?”). This task required listeners to expect pain-related cues without drawing attention to a specific source of information in the speech signal. The response prompt appeared 1000 ms after stimulus offset, and responses were recorded through a 5-button Cedrus response box ranging from “Not at all” (“Pas du tout”) on the leftmost button to “Very much” (“Énormément”) on the rightmost button. The trial ended after a response from the participant or 5 s after the question appeared, triggering the next trial after a 1500 ms blank screen. The experiment started with 8 practice trials to familiarize participants with the procedure and was then divided into 8 blocks of 40 trials, with a self-paced pause between each block to allow participants time to rest.

While performing the task, the electroencephalograms (EEGs) were recorded continuously from 64 Ag/AgCl electrodes using the ActiCap System (Brain Products, Germany). The vertical electrooculograms (VEOG) were recorded from above and below the right eye and the horizontal electrooculograms (HEOG) were recorded from the outer canthus of both eyes. The recordings were online referenced to FCz and re-referenced offline to the bilateral mastoids. The EEGs were digitized at 500 Hz and filtered with a band-pass from 0.016 Hz to 100 Hz. The EEG task lasted approximately 30 to 40 min. After the task, participants completed a series of questionnaires including the Empathic Concern and Perspective-Taking subscales of the Interpersonal Reactivity Index (IRI), plus a novel Accent-based Implicit Association Test and Stereotype Content Model questionnaire assessing implicit and explicit biases towards each cultural group (Mauchand & Pell, 2022a).

2.4. ERP Analysis

EEG recordings were pre-processed using EEGLAB (Delorme & Makeig, 2004) and ERPLAB (Lopez-Calderon & Luck, 2014). The continuous EEGs were filtered using a 40-Hz low-pass and a 0.1-Hz high-pass Butterworth of the fourth order, and were manually inspected to remove excessive movement artifact, alpha activity, or amplifier saturation. The subsequent EEGs were then decomposed with Independent

¹ For a complete list of the materials, see Mauchand & Pell, 2022b.

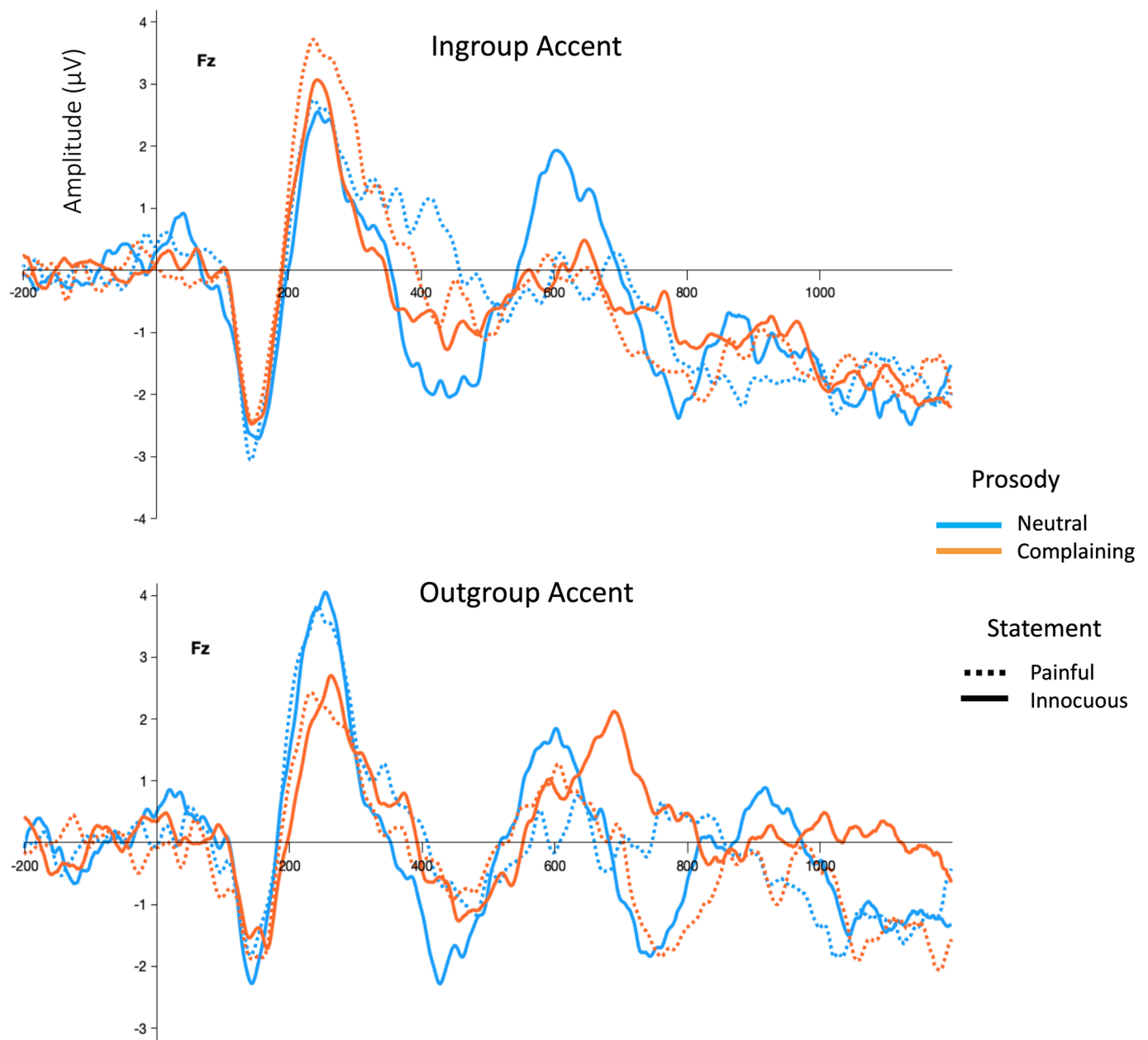


Fig. 1. Grand Average ERP waveform time-locked to the onset of the final word of utterances (RIDE-corrected). Top: Responses to utterances produced by Ingroup speakers. Bottom: Responses to utterances produced by Outgroup speakers.

Component Analysis (Makeig et al., 1996) to remove ocular artifacts.² Data were then segmented into 1200 ms epochs time-locked to the onset of the final word to explore how verbal-semantic information about the speaker's pain was processed depending on the nature of preceding prosodic and accent information available to listeners. Epochs were baseline corrected based on the mean EEG activity during the 200 ms preceding target word onset. Segments with signal peak-to-peak voltage exceeding 100 mV within a 200-ms sliding window in steps of 100 ms were automatically rejected. In total, 88% (sd = 13%) of trials were retained for analysis. None of the subjects showed enough trial rejection (i.e., more than 40% of trials) to justify exclusion from further analysis.

² For each subject, one removed ICA component was attributed to horizontal eye movement, and one (or two in a few subjects) component was attributed to vertical eye movement. Removed components were always within the first 4 components in the decomposition. No other artifact was removed at this stage.

As trial-to-trial latency jitter creates single-trial variability that reduces component discrimination, attenuates component amplitudes, and may yield erroneously significant effects, a Residue Iteration Decomposition (RIDE) procedure was performed on the ERP data before analysis (Ouyang et al., 2016). RIDE uses the latency variability and time markers to separate ERP components into predicted component clusters with a stimulus-locked cluster and one or more central clusters with unknown latency. The stimulus-locked cluster was set to a time window of 0–400 ms, and the central cluster was set to a time window of 200–1200 ms, with the latency first estimated by Woody's method within this time window. Then, ERPs were subjected to RIDE into the component clusters associated with the latency sets; these two steps were iterated until convergence. After resynchronization of the sub-component clusters to their own latency across single trials, ERPs were reconstructed accounting for variability of latency across trials (Ouyang et al., 2015). Grand average ERP waveforms are displayed in Fig. 1.

To isolate the latent structures of critical components from ERP

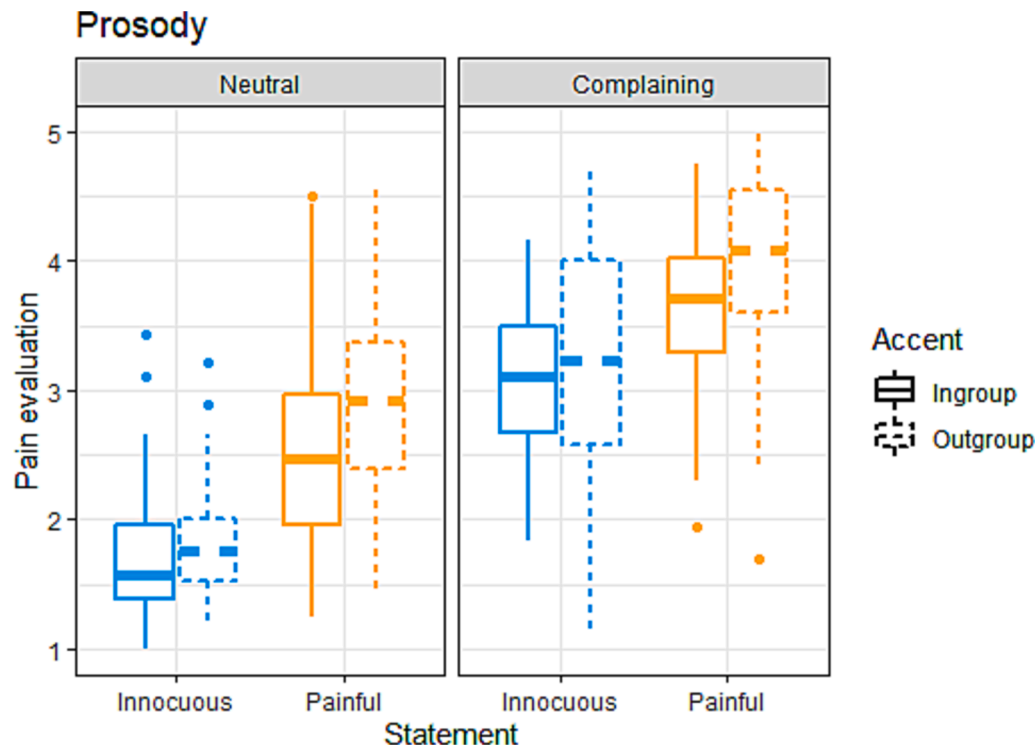


Fig. 2. Behavioral ratings of “how hurt the speaker feels” based on Prosody, Statement, and Accent. Boxplot represents the InterQuartile Range (IQR); median line represents the median rating; whiskers represent data within 1.5xIQR below or above the IQR; dots represent outlying data.

waveforms in an objective manner (Brouwer & Crocker, 2017), data were first analyzed with temporal-spatial Principal Component Analysis (PCA) (ERP PCA toolkit, see Dien, 2010). Subject-level ERP amplitudes of all channels in the entire epoch (-200 to 1200 ms) were decomposed based on their correlational structure to identify temporally and spatially distinct underlying components. The PCA was performed in two steps. First, a temporal Infomax decomposition was performed to identify temporal factors of interest; the factors to retain were determined by a Parallel Scree Test (Horn, 1965). Second, a spatial Promax rotation was performed on each surviving temporal factor to identify independent spatial component within their time window (Dien, 2012). Factors accounting for more than 0.5% of total variance were analyzed; two temporal factors, TF4 and TF5, showed significant condition effects, each across two spatial factors (Table A and Figure B1 of the Appendix). The loadings from peak channel at peak latency of each factor were rescaled to microvolts and entered into a Linear Mixed-Effect Model as responses, with Accent (Ingroup/Outgroup), Prosody (Neutral/Complaining), and Statement (Painful/Innocuous) as interacting fixed factors, and Participants as a random intercept (Loading ~ Accent*Prosody*Statement + (1|Participant)). The model was built using lme4 (Bates et al., 2015) and lmerTest (Kuznetsova et al., 2017) packages in R software (R Core Team, 2018) with Ingroup Neutral Painful utterances as the intercept. Significant effects and interactions were further explored on a by-electrode basis by entering the factors of interest together with Electrodes as an interacting nominal fixed factor (e.g., Loading ~ Accent*Prosody*Electrode + (1|Participant)), and computing pairwise comparisons at each electrode using the emmeans package in R.

3. Results

3.1. Behavioral results

A linear mixed-effect model was built with participant’s ratings as a response, Accent, Prosody and Statement as interacting fixed factors,

Participant and Item as random intercepts, and Accent, Prosody and Statement as by-Participant random slopes (Rating ~ Accent*Prosody*Statement + (1 + Accent + Prosody + Statement || Participant)). The model showed no effect of Accent ($\beta = 0.23$, $se = 0.16$, $t(17.23) = 1.40$, $p = .193$), a significant effect of Prosody ($\beta = 1.28$, $se = 0.20$, $t(17.23) = 6.52$, $p < .001$) a significant, but smaller effect of Statement ($\beta = 0.82$, $se = 0.18$, $t(13.05) = 4.59$, $p < .001$), with no interactions. Overall, speakers were rated as feeling more hurt when they produced utterances in a complaining ($M = 3.49$, $sd = 0.78$) compared to neutral ($M = 2.23$, $sd = 0.81$) prosody, and separately, when their utterances described painful ($M = 3.27$, $sd = 0.93$) versus innocuous ($M = 2.46$, $sd = 0.92$) events. Perceptual impressions of the stimuli are summarized in Fig. 2.

3.2. ERP results

3.2.1. N400

Factor TF5SF1 accounted for 7.2% of total variance in the PCA and was characterized by a large negative-going wave over the whole scalp, peaking at 442 ms at Fz, as seen in Fig. 3. The model revealed a significant effect of Statement ($\beta = -2.62$, $se = 0.60$, $t(175) = -4.41$, $p < .001$) and an interaction of Prosody and Statement ($\beta = 1.67$, $se = 0.84$, $t(175) = -1.98$, $p = .049$). Post-hoc pairwise comparisons revealed that when utterances were produced in a neutral prosody, innocuous statements yielded a significantly more negative N400 response than pain-related statements ($\beta = -2.08$, $se = 0.42$, $t(175) = -4.94$, $p < .001$). On the other hand, when produced with a complaining prosody, innocuous and painful statements did not elicit different N400 amplitudes ($\beta = -0.57$, $se = 0.42$, $t(175) = -1.35$, $p = .178$). The interaction effect was significant at electrodes AF3, AF4, AF7, AF8, Fp1, Fp2, Fz, F1, F2, F3, F4, F5, F6, F7, F8, FCz, FC1, FC2, FC3, FC4, FC5, FC6, FT7, FT8, Cz, C1, C2, C3, C4, C5, C6, T7, T8, CPz, CP1, CP2, CP3, CP4, CP5, CP6, TP8, Pz, P1, P2, P3, P4, P5, P6, P8, POz, PO3, PO4, PO7, PO8, Oz, O1, and O2. The speaker’s Accent also showed a significant main effect ($\beta = -1.62$, $se = 0.60$, $t(175) = -2.72$, $p = .007$) such that utterances spoken in an Outgroup Accent elicited a more negative response in the N400

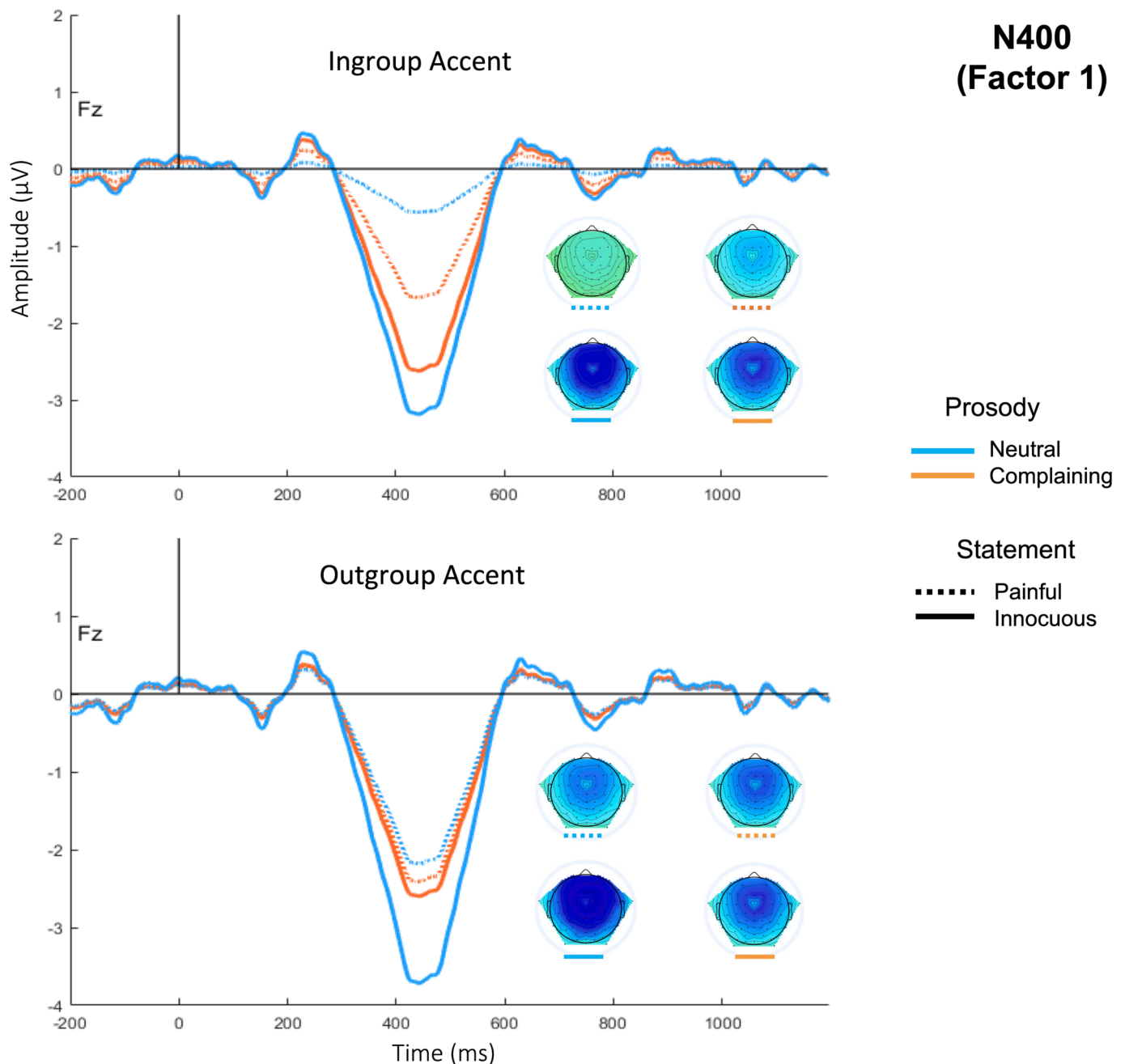


Fig. 3. ERP-PCA waves and topographic maps showing the effects of Accent, Prosody, and Statement on factor TF5SF1 (N400) at electrode Fz. Topographic maps represent voltage values across the scalp at the peak of the component (442 ms). Top: Prosody and Statement effects for utterances produced by Ingroup speakers. Bottom: Prosody and Statement effects for utterances produced by Outgroup speakers.

time window.

A second spatial component in the same temporal window (TF5SF2) was observed at right posterior electrodes, peaking at PO4 and accounting for 1.2% of total variance (see Appendix Figure C1). Analysis revealed a significant effect of Statement similar to the first N400 factor, i.e. increased negativity for innocuous vs. painful statements, without any interaction ($\beta = -0.56$, $se = 0.24$, $t(175) = -2.28$, $p = .024$). This effect was significant at C2, CPz, CP1, CP2, CP3, CP4, CP6, P1, P2, P3, P4, P5, P6, P7, P8, POz, PO1, PO2, PO3, PO4, PO7, PO8, PO9, PO10, Oz, O1, and O2.

3.2.2. P600

Factor TF4SF1 accounted for 8.8% of total variance in the PCA and was characterized by a positive-going wave over fronto-central

electrodes, peaking at 576 ms at Cz, as seen in Fig. 4. The model revealed a significant interaction of Statement and Prosody ($\beta = -2.09$, $se = 1.45$, $t(175) = -2.04$, $p = .043$). Innocuous Statements elicited a more positive response than Painful Statements when produced in a Neutral prosody ($\beta = -1.06$, $se = 0.51$, $t(175) = -2.06$, $p = .041$), but a more negative-going response when produced in a Complaining prosody ($\beta = 1.30$, $se = 0.51$, $t(175) = -2.53$, $p = .012$). This interaction was significant at electrodes Fz, F1, F2, F3, F4, FC1, FC2, FC4, Cz, C1, C2, C4, CPz, CP1, CP2, and Pz.

A second spatial component in the P600 temporal window (TF4SF2) was more posteriorly distributed, peaking at POz and accounting for 0.8% of total variance (see Appendix Figure D1). Analysis of this factor revealed that it was sensitive to the type of Statement speakers made: Painful statements tended to evoke a more positive response overall than

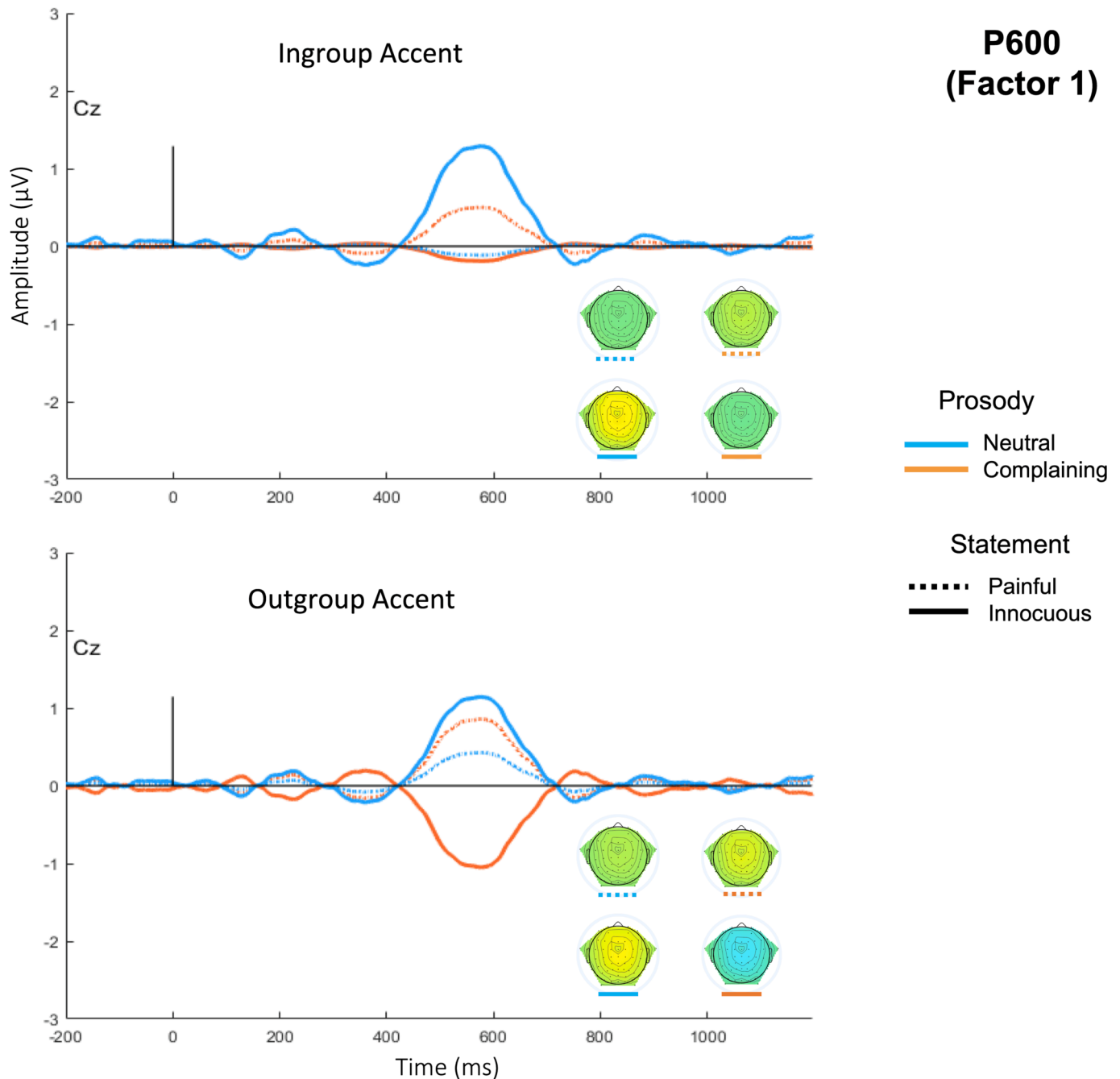


Fig. 4. ERP-PCA waves and topographic maps showing the effects of Accent, Prosody, and Statement on factor TF4SF1 (P600) at electrode Cz. Topographic maps represent voltage values across the scalp at the peak of the component (576 ms). Top: Prosody and Statement effects for utterances produced by Ingroup speakers. Bottom: Prosody and Statement effects for utterances produced by Outgroup speakers.

Innocuous Statements ($\beta = -0.72$, $se = 0.20$, $t(175) = -3.56$, $p < .001$) at posterior cortical sites (CPz, CP1, CP2, CP3, CP4, CP6, Pz, P1, P2, P3, P4, P5, P6, P7, P8, POz, PO1, PO2, PO3, PO4, PO7, PO8, PO9, PO10, Oz, O1, and O2).

4. Discussion

4.1. Effects of context on lexical retrieval

As predicted, critical words in statements characterizing seemingly harmless “innocuous” past events elicited a stronger N400 response than matched words that described obviously painful events when listeners focused on “how hurt the speaker feels”. This result highlights the

contextual sensitivity of lexical retrieval, especially regarding goal-directed speech processing (Kutas & Federmeier, 2010; Molinaro et al., 2009). In the present experiment, participants were tasked with evaluating the suffering of the speaker, thus directing their attention towards cues relevant to any form of *pain*. The utterances they heard described an action performed by a third party, with only the last word determining whether that action was harmful to the speaker or not based on the linguistic message (She said I was ... / He took my ...). It is worth noting that stimuli were not controlled for word co-occurrence, which could have influenced final word prediction via semantic association processes (Van Petten, 2014). Previous analyses of the stimulus set revealed that utterances with a pain-related final word resolution were found to be better exemplars of complaints than those with innocuous

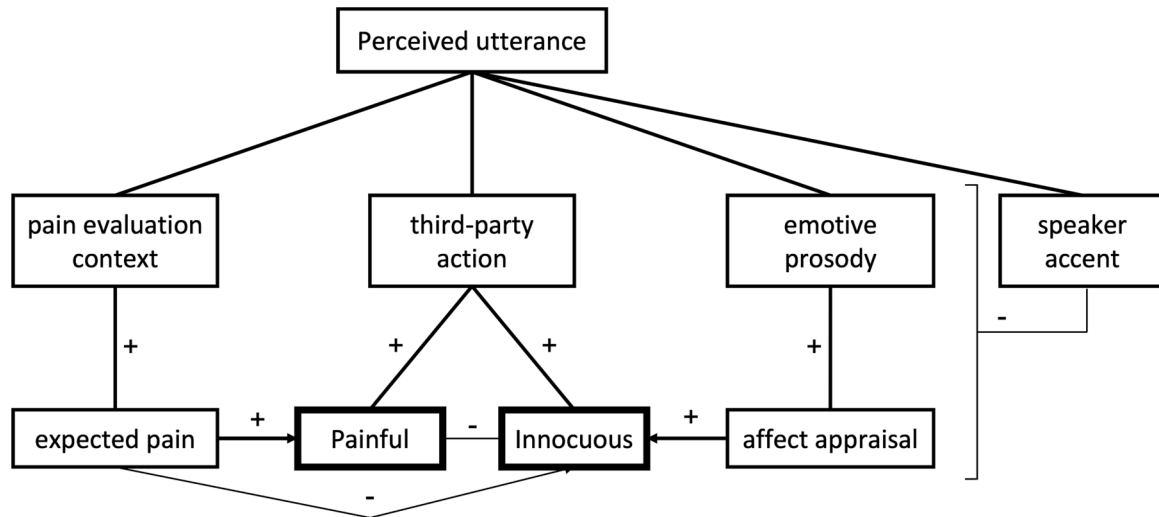


Fig. 5. Parallel constraint-satisfaction model for complaint processing. “+” indicates a constraint favoring the potential outcome, “-” indicates a constraint hindering the potential outcome.

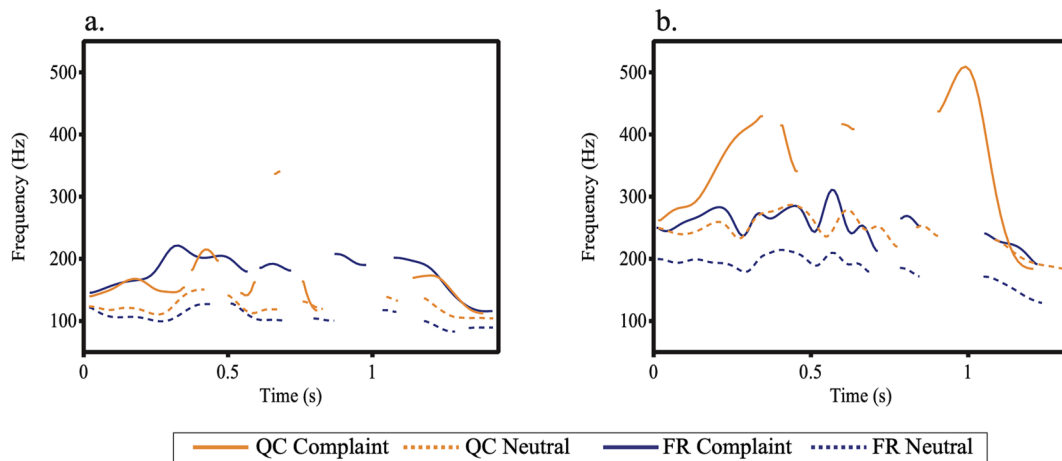


Figure A1. Example of pitch contours extracted from complaining and neutral versions of two sentences, uttered by one speaker from each group. Left: “Il a dit que j’étais stupide”/“He said I was stupid”, uttered by male speakers. Right: “Il m’ont demandé de partir”/“They asked me to leave”, uttered by female speakers.

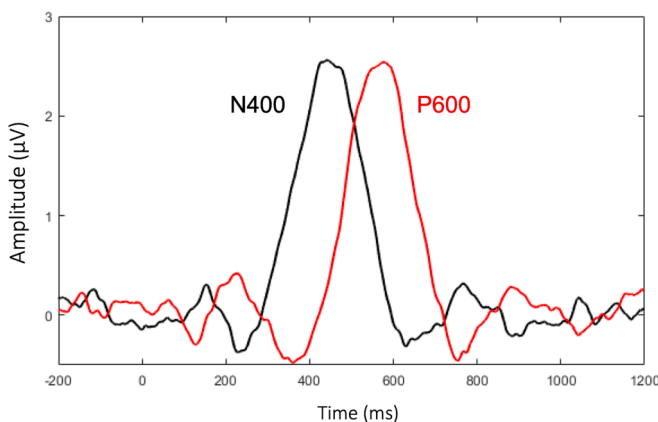


Figure B1. Temporal factor loadings for the N400 and P600 factors.

meaning (Mauchand & Pell, 2021), suggesting that the root sentence construction may indeed favor association to final words evoking pain. This sentence construction, combined with their task-induced motivated goal, would thus allow the prediction that the meaning of upcoming

words would be associated with speaker suffering. Based on our findings, it appears that words that clearly satisfied this prediction (She said I was *stupid* / He took my *spot*) were retrieved with less difficulty based on evidence in the N400 processing window, while words that described relatively innocuous events (She said I was *outside* / He took my *hand*) were associated with increased difficulties at the access stage.

These findings are in line with current language processing theories regarding lexical retrieval at the N400 (Brouwer et al., 2017; Kutas & Federmeier, 2010; Lau et al., 2008; Van Berkum et al., 2008). In particular, they highlight the dependence of the N400 on expectation (here, from task demands) and association (here, the preceding verbal information), with words that are expected and easily associated requiring less retrieval efforts (Aurnhammer et al., 2021; Delogu et al., 2019). Future investigations with tighter control over word co-occurrence may help elucidate the respective effects of semantic association, context, and pragmatic expectancies. Nevertheless, the present data extend theoretical frameworks of the N400 into a novel, socially relevant setting and provide another real-life example of how listeners make on-line predictions about upcoming words in interpersonal discourse.

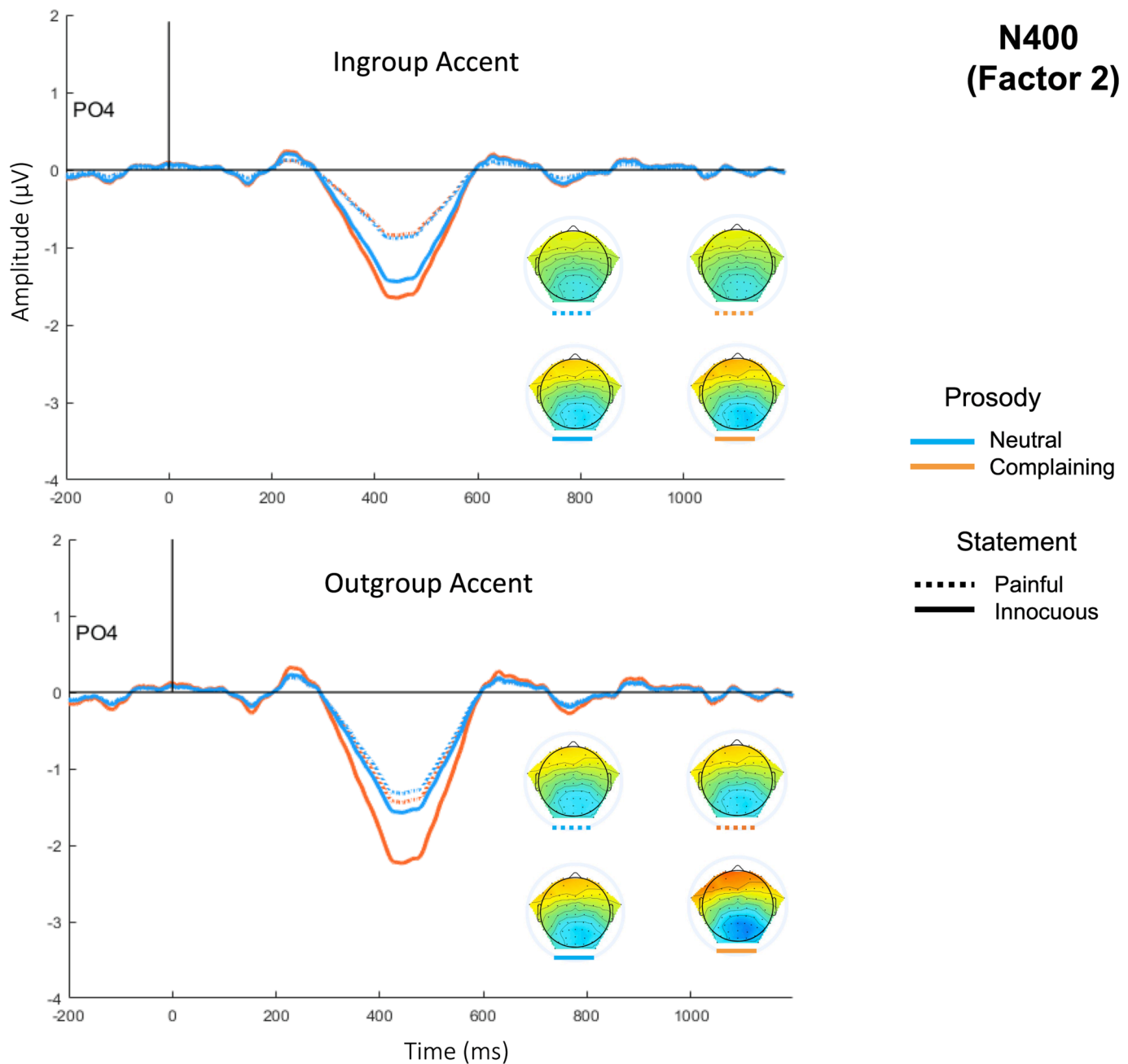


Figure C1. ERP-PCA waves and topographic maps showing the effects of Accent, Prosody, and Statement on factor TF5SF2 (N400) at electrode PO4. Topographic maps represent voltage values across the scalp at the peak of the component (442ms). Top: Prosody and Statement effects for utterances produced by Ingroup speakers. Bottom: Prosody and Statement effects for utterances produced by Outgroup speakers.

4.2. Prosodic constraints on semantic processing

While the listener's socio-evaluative goal prompted them to expect certain semantic information related to pain, a key result of this experiment is that the speaker's prosody strongly constrained this expectation at semantic processing stages. When complaints were spoken in a neutral voice, innocuous key words elicited significant increases in the N400 and P600 components when compared to painful words. However, when utterances were expressed in an affective complaining voice, there were no differences in the N400 amplitude at the critical word and the effect of the key word was reversed in the P600 window. In other words, complaining prosody, which was processed and distinguished early on from utterance onset (Mauchand & Pell, 2022), reduced retrieval efforts for words that are underspecified in a context of

pain evaluation, and altered subsequent pragmatic integration processes. As elaborated below, these effects underline the critical role of prosody during speech processing: first, as a support for a parallel constraint-satisfaction model of affective speech perception (Caffarra et al., 2019; Levy, 2008); and second, as an emotive device superseding semantic content in complaining interactions.

At the onset of the final word, listeners were aware of at least three critical aspects of the presented utterance: that it described an action performed by a third-party; that they had to evaluate how hurt the speaker felt; and that it was produced (or not) in a complaining prosody. Parallel constraint-satisfaction models posit that these pieces of information are processed in a parallel and connected manner to form impressions that will compete or converge to guide discourse interpretation (Kunda & Thagard, 1996; Levy, 2008; Pexman, 2008).

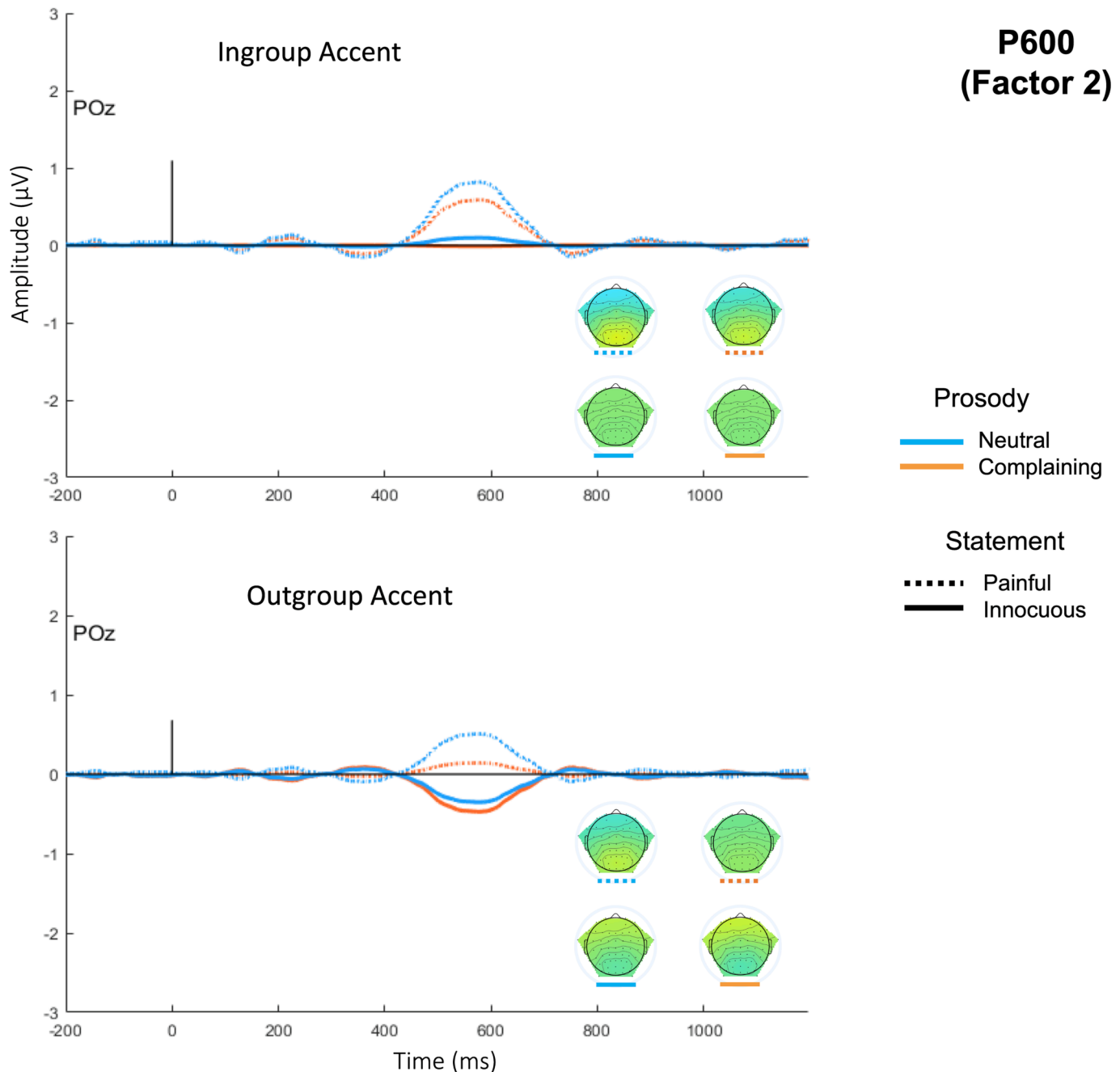


Figure D1. ERP-PCA waves and topographic maps showing the effects of Accent, Prosody, and Statement on factor TF4SF2 (P600) at electrode POz. Topographic maps represent voltage values across the scalp at the peak of the component (576ms) Top: Prosody and Statement effects for utterances produced by Ingroup speakers. Bottom: Prosody and Statement effects for utterances produced by Outgroup speakers.

Fig. 5 summarizes a model specific to complaint perception in this experiment. By knowing from preceding verbal content that the event described as a third-party action, the final word can yield two different outcomes: either the action had a harmful effect on the speaker, or it did not. As described earlier, the experimental context constrained listener's interpretation to expect a more or less harmful outcome, making words that satisfied this expectation easier to retrieve than words that did not. Prosody, on the other hand, appeared to constrain this retrieval in a different manner. Rather than narrowing listener's predictions of the upcoming word, it allowed them direct access to the speaker's affective state (Ong et al., 2018; Pell et al., 2018), thus already satisfying the task-induced prediction of suffering regardless of the outcome of the sentence. A complaining tone of voice thus facilitated the retrieval of

otherwise unexpected words, effectively reducing the N400 effect between painful and innocuous utterances.

The constraining effect of prosody continued throughout later integration efforts, albeit in a less straightforward manner, by "reversing" the P600 effect of innocuous over painful words. The P600 window often reflect pragmatic integration and updating of the meaning of utterances (Brouwer et al., 2012; Delogu et al., 2019). Negative effects have been reported in this time-window in innuendo statements, when prosody was used as a cue to interpret otherwise ambiguous statements (Rigoulot et al., 2020), suggesting a prosody-based pragmatic resolution of such ambiguity. Alternatively, the resulting increased P600 for Neutral/Innocuous and Complaint/Painful statement may reflect fine-grained interpretation of speaker intentions in unambiguous speech (i.e.,

matched valence of prosody and statement): in this case, clear evidence of the speaker's stance would allow the listener to engage in more in-depth evaluation of its pragmatic implications (Jiang et al., 2013; Jiang & Pell, 2015; Rigoulot et al., 2014). Either way, the results suggest that prosody can act as a high-constraint signal that influence N400 and P600 effects (Mauchand et al., 2021; Molinaro et al., 2009; Zhang et al., 2021), effectively superseding other contextual and verbal cues in appraising other's discourse.

The relative ease for processing innocuous complaints confirms their non-instrumental nature: one can virtually complain about anything (Alicke et al., 1992; Boxer, 1993), as long as they can efficiently reconstruct and convey their affective state (Caffi & Janney, 1994; Selting, 2010). Prosody appears as the emotive device of choice in that regard, with a clear acoustic signature that is perceptually identifiable and processed from utterance onset (Mauchand & Pell, 2021, 2022b, in review). Arguably, the necessary use of non-spontaneous speech samples here limits the "naturalness" of our stimuli and the ecological validity of the present results. Still, this study is the first to provide neurophysiological evidence that emotive prosody can take precedence over semantic information in complaining speech, complementing a sparse but growing literature in the fields of pragmatics (Drew & Walker, 2009; Edwards, 2005; Selting, 2010), acoustics (Acuña-Ferreira, 2002; Ogden, 2007; Rao, 2013), and experimental psychology (Mauchand & Pell, 2021). Behavioral results, however, suggest that this primacy is not absolute: while here, prosodic cues largely drove the perception of speaker pain, there is other evidence that perceptually hierarchize complaining topics even when spoken with an emotive voice (Alicke et al., 1992).

Put into a broader perspective, these results underline the importance of the voice as an everyday tool for coordinating meaningful social interactions (Pell & Kotz, 2021; Szczepke Reed, 2011). Beyond being a simple "readout" of one's emotions, prosody can be used to provide social information that guides a listener's processing and directs their social behavior (Scarantino, 2017; Van Kleef, 2009). By reducing processing demands on incongruously benign statements, affective prosody facilitates interpersonal affiliation even in low-involvement contexts (Boxer, 1993; Van Zant & Berger, 2020). This is exemplified by the social significance of "small talk", which is an often-underplayed yet critical aspect of our social life (Coupland, 2010). In these extraordinarily mundane interactions, prosody acts as a transverse and reliable affective signal with major implications on social outcomes (Mauchand et al., 2021; Meconi et al., 2018; Vergis et al., 2020; Zougkou et al., 2017).

4.3. Cross-cultural constraints

In addition to contextual, semantic, and prosodic information, listeners also had early access to the speaker's cultural identity through their accent. Here, accent modulated demands in the N400 processing interval, with outgroup speech eliciting larger N400 amplitudes than ingroup utterances, although these effects were independent of the speaker's prosodic expression. The accent effect shows that speaker identity information can influence processes underlying lexical access and retrieval (e.g., Foucart & Hartsuiker, 2021); however, since the speaker's accent did not interact with the prosody manipulation, our findings demonstrate that the speaker's form of vocal expression remains a robust signal in conveying social-emotional information across cultures.

In parallel-constraint satisfaction models, categorical information such as cultural identity is a critical constraint on how impressions are formed about others (Kunda & Thagard, 1996). The perception of an outgroup accent is known to affect subsequent speech processing based on intelligibility and familiarity (Derwing & Munro, 1997; Floccia et al., 2009; Jiang et al., 2020; Voeten & Levell, 2019). Related to familiarity, accented speech is often subject to biases that can alter social impressions and constrain pragmatic processing (Heblich et al., 2015; Jiang

et al., 2020; Pélissier & Ferragne, 2021). Specific to the present stimuli, for French speakers the Québécois accent is associated with implicit and explicit biases affecting its perceived status and prestige (Auger, 2005; Kircher, 2012; Mauchand & Pell, 2022a). The increase in N400 retrieval effort for outgroup utterances may thus be a consequence of such unfamiliarity and/or biases, which caused greater uncertainty about the meaning of the upcoming critical word. Previous analyses of ERP responses to complaints at utterance onset indeed revealed that outgroup vocal expressions were less motivationally relevant and required slower encoding efforts, with effects visible as early as the P200 component (Mauchand & Pell, 2022b). Interestingly, while P200 effects were not the focus of the present paper, visual inspection of grand average ERPs at this time window shows patterns analogous to those reported at utterance onset (Mauchand & Pell, 2022b); future research should further investigate these early events and how they influence later linguistic and pragmatic processes. Mauchand & Pell (2022b) suggest that listeners may be less attuned to cues about how outgroup speakers feel, which, in this case, could lead to more effortful semantic processing of the outgroup speaker message (Jiang et al., 2020). Still, the lack of interactions between accent and prosody at the N400 suggests that there must be strong overlap in how emotivity is expressed between our cultural groups (Laukka & Elfénbein, 2020; Liu et al., 2015; Thompson & Balkwill, 2006). Prosody thus appears to elicit social impressions that are consistent across regional cultures and constrain empathic speech processing in the same way. As shown in Fig. 5, accent constrains how the speaker's discourse is appraised in general, regardless of other vocal information.

5. Conclusion

The present study and results provide neurophysiological evidence for a parallel constraint-satisfaction model of affective speech processing. Extending neurolinguistic literature on the N400 and P600 to the social speech act of complaining, it suggests that expectation- and association-based retrieval efforts can be mitigated by parallel information from relevant *para*-linguistic signals such as speech prosody and accent. In the case of complaints, these results confirm that a speaker can virtually complain about anything, to anyone, as long as they provide ostensive evidence of their emotive disposition. More generally, our data emphasize the relevance of prosody in facilitating empathic and affiliative mechanisms in social interactions, suggesting that the "how" (and the "who") often supersedes the "what" in emotive speech communication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Mauchand, Maël; Pell, Marc D (2023), "ERP study on the empathic processing of complaining speech", Mendeley Data, V3, doi: 10.17632/zdgyx5mf9t.3

Acknowledgements

This research was supported by an Insight Grant to MP from the Social Sciences and Humanities Research Council of Canada (G245592 SSHRC 435-2017-0885) as well as scholarships to MM from McGill Faculty of Medicine, Fonds de Recherche du Québec - Nature et Technologie (FRQNT), and the Quebec Bio-Imaging Network (QBIN/RBIQ). Data and codes used for this experiment are available on Mendeley Data.

Appendix

See

Table A
Summary of PCA temporal-spatial factors accounting for over 0.5% of total variance. Bolded rows indicate factors that showed significant condition effects.

Factor	peak latency (ms)	negative peak	positive peak	variance explained
TF1SF1	1036	EOG2	–	12.70%
TF1SF2	1036	EOG2	EOG1	6.10%
TF1SF3	1036	EOG2	POz	3.30%
TF1SF4	1036	F7	EOG2	1.80%
TF2SF1	814	–	Fz	8.50%
TF2SF2	814	EOG2	EOG1	1.10%
TF2SF3	814	Fp1	P2	0.50%
TF3SF1	690	Fz	–	8.10%
TF3SF2	690	Fp1	POz	0.80%
TF4SF1	576		Cz	8.80%
TF4SF2	576	Fp1	POz	0.80%
TF5SF1	442	Fz	–	7.20%
TF5SF2	442	PO4	Fp1	1.20%
TF5SF3	442	EOG2	FT10	0.70%
TF6SF1	238	–	Cz	4.70%
TF6SF2	238	P2	F8	0.70%
TF7SF1	936	–	Fz	1.60%
TF8SF1	144	Fz	–	0.90%

Figs. A1–D1).

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